

Positive Square Energy for Two C_5 Blocks Joined by a Bridge

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Abstract

Let H be the ten-vertex graph formed from two disjoint copies of C_5 by joining one vertex of each copy with a bridge. We prove that attaching arbitrary trees at any vertices of this core produces a graph G satisfying

$$s^+(G) \geq |V(G)| + s^+(H) - 10 > |V(G)|.$$

The proof eliminates the trees by matching belief propagation and compares the continuous Coulson phase with that of H . The angular comparison is an exact coefficientwise certificate with 4891 positive nonconstant terms. We also factor the characteristic polynomial of H and evaluate $s^+(H)$ exactly.

1 Statement and notation

All graphs are finite, simple, and undirected. For adjacency eigenvalues $\lambda_1, \dots, \lambda_n$, write

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2.$$

Let H consist of two vertex-disjoint 5-cycles and exactly one edge joining one vertex of the first cycle to one vertex of the second. Thus H has ten vertices and eleven edges.

Theorem 1.1. *Suppose that G is obtained from H by attaching arbitrary rooted trees at arbitrary core vertices (identifying each tree root with its chosen core vertex). If $n = |V(G)|$, then*

$$s^+(G) \geq n + s^+(H) - 10 > n. \tag{1}$$

More precisely,

$$s^+(H) = \frac{31 + \sqrt{13}}{2} - \sqrt{5} - \rho^2, \tag{2}$$

where ρ is the unique root in $(2, \sqrt{5})$ of

$$\rho^3 - 4\rho - 1 = 0. \tag{3}$$

The sign in (3) is worth emphasizing. The characteristic polynomial has the factor $x^3 - 4x + 1$. Its negative root is $-\rho$, and substitution of $x = -\rho$ gives $\rho^3 - 4\rho - 1 = 0$. In particular, $x^3 - 4x + 1$ has no root in $(2, \sqrt{5})$; its largest positive root is less than 2. Confusing that positive root with the magnitude of the negative root changes the sign in (3) and makes the interval assertion false.

2 Tree elimination by matching messages

For a forest F , define its signless matching partition

$$Z_F(t) = \sum_M t^{|V(F)| - 2|M|}, \quad (4)$$

where the sum is over all matchings. For a graph with positive vertex activities $a = (a_v)$, use the weighted version

$$Z_F(a) = \sum_M \prod_{v \text{ unmatched by } M} a_v. \quad (5)$$

All edge activities are one.

Orient every branch outside the core toward the core. If $u \rightarrow v$ is an oriented branch edge, let $T_{u \rightarrow v}$ be the component below u , including u , and put

$$m_{u \rightarrow v}(t) = \frac{Z_{T_{u \rightarrow v}}(t)}{Z_{T_{u \rightarrow v} - u}(t)}. \quad (6)$$

Lemma 2.1 (Matching belief propagation). *The messages satisfy*

$$m_{u \rightarrow v}(t) = t + \sum_{w \text{ child of } u} \frac{1}{m_{w \rightarrow u}(t)} \geq t > 0. \quad (7)$$

After all branches are eliminated, each core vertex has activity

$$a_v(t) = t + \sum_{u \text{ attached at } v} \frac{1}{m_{u \rightarrow v}(t)} = t + y_v(t), \quad y_v(t) \geq 0. \quad (8)$$

Every term in the grouped Sachs expansion has the same positive real factor

$$K(t) = \prod_{u \text{ adjacent to the core}} Z_{T_{u \rightarrow v}}(t) > 0. \quad (9)$$

This remains true for a Sachs term in which either entire pentagon is deleted.

Proof. Split a matching below u according to whether u is unmatched or is matched to one of its children, and factor the child-subtree partitions. This proves (7); induction from a leaf gives the inequality. For a branch rooted next to a core vertex, extracting its full partition changes the weight of using the edge into the branch to $Z_{T-u}/Z_T = 1/m_{u \rightarrow v}$. The mutually exclusive choices over incident branches give (8), while independence gives (9).

If a core vertex is deleted as part of a Sachs cycle, every branch attached there becomes a disconnected full tree and still contributes its factor Z_T to $K(t)$. Retained core vertices have the same effective activities as before. Hence deleting a whole pentagon does not leave an additional ratio of branch partitions. $\square$

3 The weighted bridge core

Let

$$\phi_G(z) = \det(zI - A(G)), \quad \Psi_G(t) = i^{-n} \phi_G(it) = \prod_{j=1}^n (t + i\lambda_j), \quad t > 0.$$

In the grouped Sachs expansion, a 5-cycle has multiplier $-2i^{-5} = 2i$. The two cycles are disjoint, so their joint selection has multiplier $(2i)^2 = -4$.

Root one weighted pentagon at its bridge endpoint and list its other activities consecutively as x_1, x_2, x_3, x_4 . Define

$$P(x_1, x_2, x_3, x_4) = x_1x_2x_3x_4 + x_3x_4 + x_1x_4 + x_1x_2 + 1, \quad (10)$$

$$B(x_1, x_2, x_3, x_4) = x_2x_3x_4 + x_2 + x_4 + x_1x_2x_3 + x_1 + x_3. \quad (11)$$

Here P is the matching partition after deleting the root. If the root activity is a_0 , the full pentagon partition is

$$Q = a_0P + B. \quad (12)$$

For the left pentagon denote the deleted-root and full partitions by p, q ; for the right one denote them by s, r , respectively.

Proposition 3.1 (Exact bridge formula). *With the activities (8),*

$$\boxed{\frac{\Psi_G(t)}{K(t)} = (qr + ps - 4) + 2i(q + r).} \quad (13)$$

Proof. Splitting matchings according to whether the bridge is unused or used gives $qr + ps$: in the second case both bridge endpoints are deleted. Selecting the left Sachs cycle leaves the full right pentagon and contributes $2ir$; selecting the right one contributes $2iq$. Selecting both contributes -4 . Lemma 2.1 supplies the common factor $K(t)$. $\square$

For bare activities, set

$$P_0 = t^4 + 3t^2 + 1, \quad Q_0 = t^5 + 5t^3 + 5t. \quad (14)$$

Introduce the real and imaginary coordinates

$$\begin{aligned} X &= qr + ps - 4, & Y &= 2(q + r), \\ X_0 &= Q_0^2 + P_0^2 - 4, & Y_0 &= 4Q_0. \end{aligned}$$

Both Y and Y_0 are positive for $t > 0$. We therefore use the upper-half-plane branch

$$\Theta_G(t) = \text{atan2}(Y, X), \quad \Theta_H(t) = \text{atan2}(Y_0, X_0), \quad 0 < \Theta_G, \Theta_H < \pi. \quad (15)$$

In particular,

$$\Theta_H(t) = \text{atan2}(4Q_0, Q_0^2 + P_0^2 - 4). \quad (16)$$

An ordinary one-argument arctangent cannot replace this expression globally:

$$X_0 = t^{10} + 11t^8 + 41t^6 + 61t^4 + 31t^2 - 3$$

is negative for all sufficiently small positive t .

4 Coefficientwise angular certificate

Proposition 4.1. *For all activities $a_v = t + y_v$ with $t > 0$ and $y_v \geq 0$,*

$$Y_0 X - Y X_0 \geq 0. \quad (17)$$

Consequently,

$$0 < \Theta_G(t) \leq \Theta_H(t) < \pi. \quad (18)$$

The cross-product inequality is an equality at the bare specialization $y_0 = \dots = y_9 = 0$.

Proof. After canceling a positive factor 2, inequality (17) is

$$2Q_0(qr + ps - 4) - (q + r)(Q_0^2 + P_0^2 - 4) \geq 0. \quad (19)$$

Substitute $a_v = t + y_v$ at the ten core vertices, use (10)–(12), and expand over $\mathbb{Z}$ in the ordered variables $(t, y_0, \dots, y_9)$. The exact expansion has 4891 nonzero monomials. Every coefficient is positive, the minimum is 1, the maximum is 289, and every monomial contains at least one y_v . Thus the polynomial is nonnegative on the nonnegative orthant and has zero bare specialization. The exhaustive symbolic verification and its canonical digest are recorded in Appendix A.

It remains to translate the cross product without losing the quadrant. From (15), the difference $\Theta_G - \Theta_H$ belongs to $(-\pi, \pi)$, and

$$\sin(\Theta_G - \Theta_H) = \frac{Y X_0 - X Y_0}{\sqrt{X^2 + Y^2} \sqrt{X_0^2 + Y_0^2}} \leq 0.$$

On $(-\pi, \pi)$ this implies $\Theta_G - \Theta_H \leq 0$. No sign assumption on X or X_0 and no ordinary-arctangent division is used. $\square$

5 Phase integration and the bare spectrum

For the continuous argument in (15), the signed square-energy Coulson identity is

$$s^+(F) - s^-(F) = -\frac{4}{\pi} \int_0^\infty t \Theta_F(t) dt. \quad (20)$$

It follows by applying the scalar identity to the factors $t + i\lambda_j$; the trace relation $\sum_j \lambda_j = 0$ cancels the nonintegrable first-order term at infinity. Proposition 4.1 and (20) give

$$s^+(G) - s^-(G) \geq s^+(H) - s^-(H). \quad (21)$$

We now evaluate the right side exactly. A direct determinant calculation gives

$$\phi_H(x) = (x - 1)(x^2 - x - 3)(x^2 + x - 1)^2(x^3 - 4x + 1). \quad (22)$$

The cubic has two positive roots α, β and one negative root $-\rho$. Indeed, $f(x) = x^3 - 4x + 1$ has one root in each of $(-3, -2)$, $(0, 1)$, and $(1, 2)$. Vieta's relations give

$$\alpha + \beta = \rho, \quad \alpha\beta = \rho^2 - 4, \quad \alpha^2 + \beta^2 = 8 - \rho^2.$$

The other positive eigenvalues read from (22) are

$$1, \quad \frac{1 + \sqrt{13}}{2}, \quad \frac{-1 + \sqrt{5}}{2} \quad \text{with multiplicity two.}$$

Squaring and summing proves (2).

For completeness, the definition of ρ and the strict slack can be verified without decimals. Let $g(x) = x^3 - 4x - 1$. Then

$$g(2) = -1 < 0, \quad g(\sqrt{5}) = \sqrt{5} - 1 > 0, \quad g'(x) = 3x^2 - 4 > 0 \quad (x > 2).$$

Hence g has exactly one root ρ in $(2, \sqrt{5})$, and $\rho^2 < 5$. Therefore

$$s^+(H) > \frac{21 + \sqrt{13}}{2} - \sqrt{5} > 10. \quad (23)$$

The last inequality is exact: it is equivalent to $(1 + \sqrt{13})/2 > \sqrt{5}$, and both sides are positive; after moving 1 and squaring it reduces to $13 > (2\sqrt{5} - 1)^2 = 21 - 4\sqrt{5}$, equivalently $\sqrt{5} > 2$.

Finally, G is connected and bicyclic, so it has $n + 1$ edges. Thus

$$s^+(G) + s^-(G) = \text{tr } A(G)^2 = 2n + 2, \quad s^+(H) + s^-(H) = 22. \quad (24)$$

Averaging (21) with (24) yields

$$s^+(G) \geq n + 1 + \frac{s^+(H) - s^-(H)}{2} = n + s^+(H) - 10.$$

The strict inequality in (1) follows from (23). Equivalently, the exact bare phase integral is

$$\int_0^\infty t \Theta_H(t) dt = \frac{\pi}{2} (11 - s^+(H)) = \frac{\pi}{2} \left(\rho^2 + \sqrt{5} - \frac{9 + \sqrt{13}}{2} \right) < \frac{\pi}{2}.$$

A Exact certificate and reproducibility

The certificate script is

```
positive-square-energy/experiments/c5_bridge_phase_certificate.py.
```

Run it from the repository root with

```
python positive-square-energy/experiments/c5_bridge_phase_certificate.py
```

It constructs (19) over $\mathbb{Z}$, checks the bare characteristic-polynomial factorization (22), and asserts all of the following for the coefficient expansion:

1. there are exactly 4891 nonzero monomials;
2. every nonzero coefficient is positive, with minimum 1 and maximum 289;
3. every monomial contains at least one of $y_0, \dots, y_9$;
4. the bare specialization is zero; and
5. the SHA-256 digest of the canonical ordered stream `monomial:coefficient`, one term per line, is

```
365e18dedf9032fbfdb88af83d033f0651f02412c796b4f1dfde04152a478af1.
```

The digest commits to the complete coefficient list rather than to a numerical sampling of activities.